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Risk Management in Border Territories

Wildfires, critical infrastructure and the population
El Tarf, Algeria · Jendouba, Tunisia

Abderaouf CHEKKOUR and Mohamed Mounir MESSEGUEM | Master 2, Regional Planning (2025)

Research question

How does geographic continuity across the Algeria–Tunisia border relate to the exposure of roads, electricity networks and communities to wildfire?

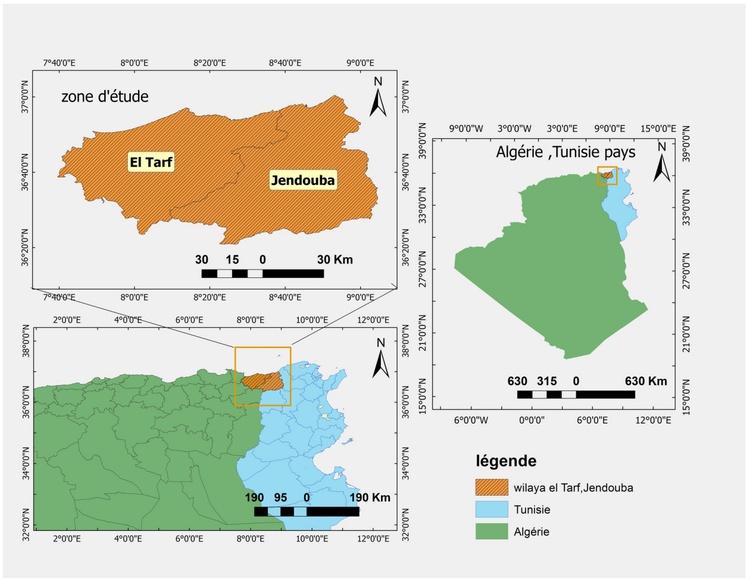


Figure 1. Location of the El Tarf–Jendouba study area.

Fire history reported in the original poster

Area	Year	Dates	Deaths	Burned area
Jendouba	2017	Jul 29–Aug 7	0	1,412 ha
Jendouba	2022	Aug 17–18	0	1,923 ha
El Tarf	2017	Jul 29–Aug 7	0	9,077 ha
El Tarf	2022	Aug 17–18	36	11,396 ha

Figure 2. Original reported values; not recalculated or independently verified. The original gives 2017 and 2022 events the same dates in both regions; source-level checking is still needed.

Data and methods

Satellite imagery and thematic cartography in ArcGIS Pro; remote sensing using NDVI and NBR. The source poster lists Sentinel-2A imagery at 10 m and 20 m, NASA FIRMS, OpenStreetMap and the ENTSO-E 2023 interconnected network map.

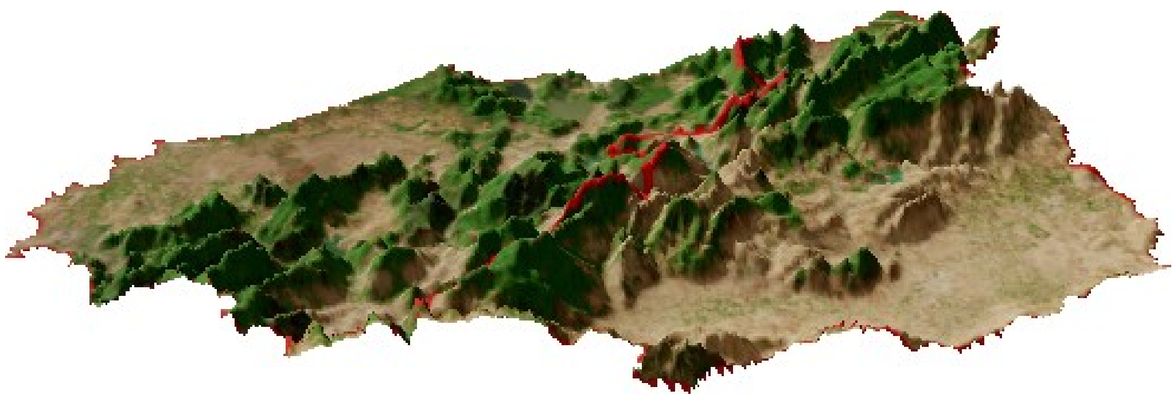


Figure 3. Original three-dimensional view of regional topography.

Shared landscape and cross-border planning

The maps place terrain, vegetation, settlements and infrastructure in a shared border landscape. They support discussion of coordinated preparedness and evacuation planning. Landscape continuity alone does not establish that a particular fire crossed the international border.

Critical infrastructure and population

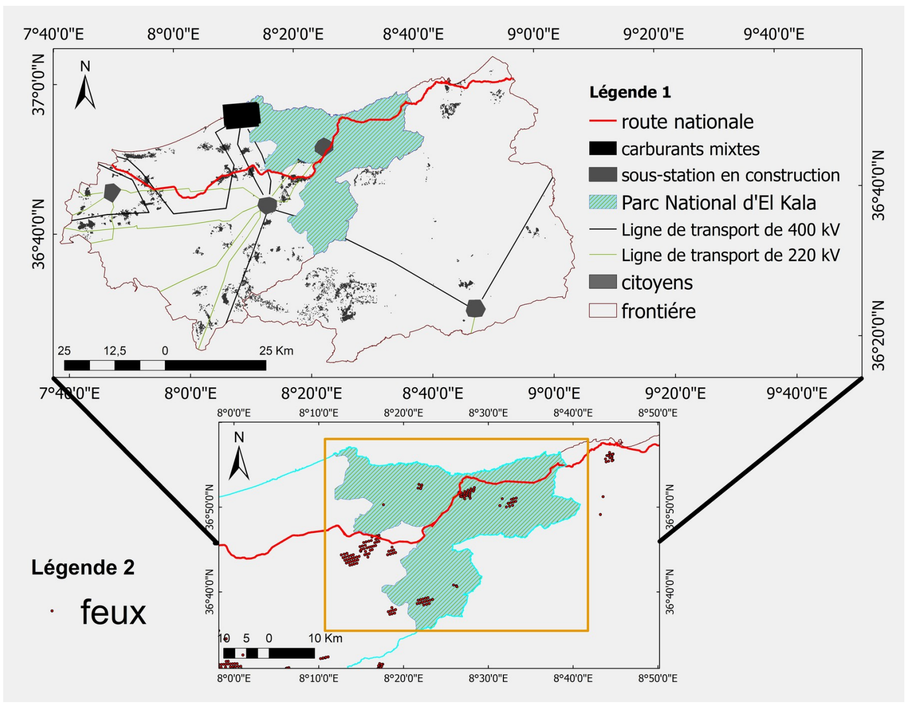


Figure 4. Original infrastructure and fire-context maps for August 2022. The poster discusses National Road 44 and a 400 kV electricity interconnection.

The poster examines how wildfire exposure can affect road access, electricity infrastructure and communities. Fire-related damage and smoke-related visibility problems are distinct mechanisms; the maps do not by themselves verify a specific cable failure, road closure or service interruption.

Fire detections and vegetation index

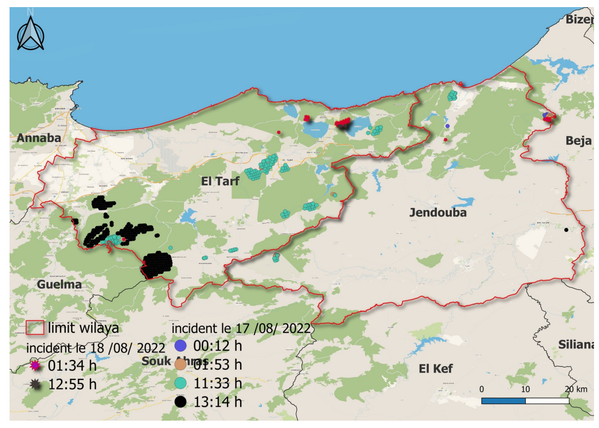


Figure 5. Original mapped fire detections and observation times. Detection times are not established ignition times.

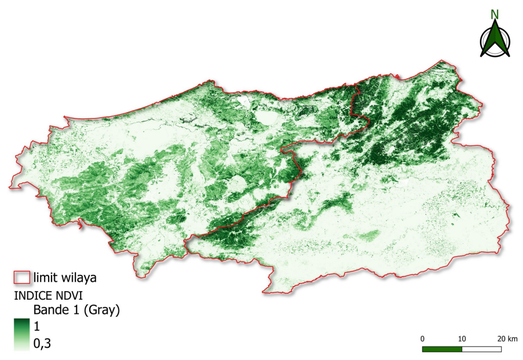


Figure 6. Original NDVI map. NDVI is a spectral index, not a direct measure of vegetation density in every setting.

Interpretation and limits

Similar observation periods across the two regions may be consistent with shared weather conditions, but do not prove common ignition causes or climate-change attribution. The original poster discusses severe heat and drought; a heat-dome mechanism and temperatures above 47°C require dated meteorological evidence.

The original reports a greater burned area in El Tarf than in Jendouba. These values and the reported fatalities need traceable, event-specific references. The contribution is an academic poster, not a validated operational risk forecast.

Map labels and legend key

Original map labels remain in French. Route nationale: national road; ligne de transport: transmission line; sous-station en construction: substation under construction; frontière: border; feux: fires; limite wilaya: provincial boundary. The labels “carburants mixtes” and “citoyens” need clarification in the source GIS project.

Provenance and sources

English adaptation, October 29, 2026. Original maps, co-authorship and reported values retained; duplicated conclusion removed and causal claims qualified. No new calculations. Original source list: Sentinel-2A; NASA FIRMS; OpenStreetMap; ENTSO-E 2023; El Moudjahid; Dzair Tube. Source poster gives news homepages rather than exact articles.